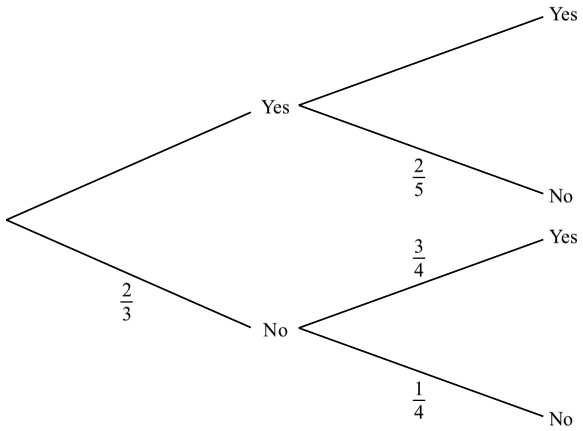
Question	Answer	Marks	Partial Marks
1	–2, –1, 0, 1	**2**	**B1** if one error/omission/extra
2	Correct triangle with arcs	**2**	**B1** correct no arcs
3	x^9y^4 final answer	**2**	**B1** for x^9y^k or x^ky^4 or correct answer seen then spoiled
4	–14, –5, 1	**2**	**B1** for 2 correct
5(a)	32.7 to 33.3	**1**	**FT** *their* printed diagram
5(b)	118 to 120	**1**	**FT** *their* printed diagram
5(c)	298 to 300	**2**	**B1** for 360 – (180 – *their* **(b)**) **FT** *their* **(b)** + 180
6(a)	5	**2**	**M1** for 10 × 12 × 'h' = 600
6(b)	172 or 172.4 to 172.5	**2**	**M1** for $1120 \times \frac{4}{3} \times \pi \times 0.45^3$
7(a)	20	**1**	
7(b)	 $\frac{2}{3}, \frac{2}{5}, \frac{3}{4}, \frac{1}{4}$ oe	**2**	**M1** for 2 or 3 correct
7(c)	$\frac{7}{10}$ oe	**3**	**M2** for $\frac{1}{3} \times \frac{3}{5}$ + *their* $\frac{2}{3}$ × *their* $\frac{3}{4}$ or **M1** for one product

© Cambridge University Press & Assessment 2022

Question	Answer	Marks	Partial Marks
19	391 or 391.1 to 391.2	4	**M3** for $3 \times 6 \times 20 + 2 \times \frac{1}{2} \times 6 \times 6 \times \sin 60$ oe OR **M1** for [3 ×] 6×20 **M1** for [2 ×] $\frac{1}{2} \times 6 \times 6 \times \sin 60$ oe
20	[k =] 9 [u =] 3	2	**B1** each
21	$\frac{5p}{2(p+4)}$ or $\frac{5p}{2p+8}$ final answer	3	**B1** for $5p(p-4)$ **B1** for $2(p+4)(p-4)$
22(a)(i)	$-\frac{1}{3}\mathbf{p} + \frac{1}{3}\mathbf{t}$ or $\frac{1}{3}(-\mathbf{p} + \mathbf{t})$ final answer	2	**M1** for $\overrightarrow{PT} = -\mathbf{p} + \mathbf{t}$ or $\overrightarrow{TP} = -\mathbf{t} + \mathbf{p}$
22(a)(ii)	$\frac{2}{3}\mathbf{p} - \frac{1}{3}\mathbf{t}$ or $\frac{1}{3}(2\mathbf{p} - \mathbf{t})$ final answer	2	**M1** for a correct route
22(b)	$\overrightarrow{OM} = \frac{2}{3}\mathbf{t} - \frac{2}{3}\mathbf{t} + \frac{4}{3}\mathbf{p}$ oe	M1	
	$\overrightarrow{OM} = \frac{4}{3}\mathbf{p}$ and $\overrightarrow{OM}$ is a multiple of $\overrightarrow{OP}$ oe	A1	
23	3.6	3	**M1** for $\frac{\textit{their}\text{ distance}}{\textit{their}\text{ time}}$ **B1** for 7.95 or 7950 or 132.5

© Cambridge University Press & Assessment 2022